

SAMMER HUSSAIN

AI Automation Engineer

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SUMMARY

AI Automation Engineer with hands-on experience building production-ready Generative AI and automation systems. Skilled in developing RAG-based applications, AI-powered products, and end-to-end automation workflows using n8n. Experienced in integrating LLMs, vector databases, and cloud tools to automate document processing, lead enrichment, and intelligent data pipelines. Focused on turning unstructured data and manual processes into scalable, reliable AI-driven solutions.

EXPERIENCE

AI Engineer Intern

CoreTech Innovations

- 08/2025 - 09/2025 Hyderabad, Sindh, Pakistan
- Developed ML models for sentiment analysis and student marks prediction using Python, Scikit-learn, and TensorFlow.
 - Built a smart service platform to streamline workflows and boost productivity.
 - Handled data preprocessing, feature engineering, model evaluation, and optimization for production-ready solutions.
 - Collaborated in Agile teams and communicated insights to drive data-informed business decisions.

EDUCATION

Bachelor of Science in Information Technology

Quaid-e-Awam University of Engineering, Sciences and Technology

12/2021 - 09/2025 Nawabshah, Pakistan

CGPA
3.58 / 4.0

SKILLS

Python	JavaScript	SQL	N8N	LangChain	RAG
Prompt Engineering	ChromaDB	Pinecone	Hugging Face		
OpenAI	Google Gemini	Transformers	Model fine-tuning		
NLTK	Scikit-learn	TensorFlow	Keras	OpenCV	
Flask	Streamlit	Node JS	React JS	SQLite	
PostgreSQL	MongoDB	Git	GitHub		
Visual Studio Code	Anaconda	Jupyter Notebook			
Google Colab	Docker (Basic)	AWS (Basic)			

CERTIFICATION

- | | |
|-----------------------------|-------------------------------|
| Google AI Essentials | Introduction to Generative AI |
| Google Prompting Essentials | |

PROJECTS

FredCoach - AI Coaching Platform

- 10/2025 - 11/2025
- Full-Stack AI Coaching SaaS Platform
- Full-stack SaaS delivering 24/7 AI mental coaching via GPT-4o chatbot, leveraging RAG with ChromaDB as the vector database.
 - Built with Node.js, Express, MongoDB, React, and Tailwind CSS.
 - Integrated Stripe for secure subscription management and developed a comprehensive admin panel for user management, analytics, and platform customization.

Content Examiner - AI-Powered Plagiarism Detection System (FYP)

- 02/2025 - 06/2025
- Final year project focused on plagiarism detection.
- Modular Flask + React app detecting semantic & lexical plagiarism using TF-IDF, cosine similarity, and RoBERTa for sentence-level semantic analysis.
 - Supports document uploads, AI-generated text detection, and PDF reporting with optimized NLP preprocessing (tokenization, lemmatization, stop-word removal) to enhance accuracy and reduce analysis noise.

Document Ingestion and Vectorization Pipeline

- 01/2026 - 01/2026
- Document Ingestion and Vectorization Pipeline using n8n, Google Gemini, Pinecone.
- End-to-end n8n workflow fetching files from Google Drive, embedding with Google Gemini, and indexing in Pinecone for semantic search and RAG applications.
 - Scalable automation integrating LLMs, vector stores, and cloud storage.

Live Object Detection with YOLOv8

- 08/2024 - 09/2024
- Live Object Detection with YOLOv8
- Real-time multi-class object detection with Flask, OpenCV, YOLOv8.
 - Features webcam feed, live predictions, bounding boxes, confidence scoring, and low-latency inference in a web interface.

LinkedIn Lead Outreach Automation

- 12/2025 - 12/2025
- LinkedIn Lead Outreach Automation using n8n, LangChain, Google Gemini, and Google Sheets.
- Built AI driven business automation workflow generating personalized outreach emails from Google Sheets data.
 - Integrated LangChain with Gemini for intelligent email generation based on lead profiles and business context outreach.